DATABASE TECHNOLOGIES LAB SHEET – 10

Course Code: MCA105P
Topic: Hospital Medicine Management System Using MySQL

Q1. Create Database and Insert Medicine Records

```
CREATE DATABASE HospitalDB;
USE HospitalDB;

CREATE TABLE medicines (
   medicine_id INT PRIMARY KEY,
   name VARCHAR(100),
   category VARCHAR(50),
   price DECIMAL(10,2),
   stock INT,
   manufacturer VARCHAR(100)
);

INSERT INTO medicines VALUES
(301, 'Paracetamol', 'Analgesic', 25, 100, 'ABC Pharma'),
(302, 'Amoxicillin', 'Antibiotic', 80, 50, 'HealthCare Ltd'),
(303, 'Cetirizine', 'Antihistamine', 35, 20, 'ABC Pharma'),
(304, 'Omeprazole', 'Antacid', 60, 15, 'MediLife'),
(305, 'Azithromycin', 'Antibiotic', 120, 40, 'HealthCare Ltd'),
(306, 'Vitamin C', 'Supplement', 45, 80, 'Wellness Pharma'),
(307, 'Ibuprofen', 'Analgesic', 90, 25, 'MediLife'),
(308, 'Calcium', 'Supplement', 150, 60, 'Wellness Pharma');

SELECT * FROM medicines;
```

Q2. Create Medicine Entry Table

```
CREATE TABLE medicine_entry (
   medicine_id INT PRIMARY KEY,
   medicine_name VARCHAR(100) NOT NULL,
   category VARCHAR(50),
   price DECIMAL(10,2),
   stock INT,
   manufacturer VARCHAR(100)
);

DESC medicine_entry;
```

Q3. Apply Design and Constraints to Medicine Table

```
ALTER TABLE medicine_entry
MODIFY medicine_name VARCHAR(100) NOT NULL;

ALTER TABLE medicine_entry
MODIFY category VARCHAR(50) NOT NULL;

ALTER TABLE medicine_entry
MODIFY price DECIMAL(10,2) NOT NULL;

ALTER TABLE medicine_entry
MODIFY stock INT NOT NULL;

ALTER TABLE medicine_entry
MODIFY manufacturer VARCHAR(100) NOT NULL;

DESC medicine_entry;
```

Q4. Connect to HospitalDB

```
SHOW DATABASES;

USE HospitalDB;

SHOW TABLES;

DESC medicines;

SELECT * FROM medicines;
```

Q5. Insert New Medicine Record

```
INSERT INTO medicines
VALUES (309, 'Dolo 650', 'Analgesic', 30, 100, 'ABC Pharma');

SELECT * FROM medicines;
```

Q6. Retrieve and Display All Medicines

```
SELECT * FROM medicines;

SELECT medicine_id, name, category, price, stock, manufacturer
FROM medicines;
```

Q7. Search Medicine by ID or Name

```
SELECT * FROM medicines
WHERE medicine_id = 301;

SELECT * FROM medicines
WHERE name = 'Paracetamol';

SELECT * FROM medicines
WHERE name LIKE '%Vitamin%';
```

Q8. Update Medicine Stock

```
UPDATE medicines
SET stock = 150
WHERE medicine_id = 301;

SELECT * FROM medicines
WHERE medicine_id = 301;
```

Q9. Delete Medicine Using Medicine ID

```
DELETE FROM medicines
WHERE medicine_id = 308;

SELECT * FROM medicines;
```

Q10. MySQL Query Practice

(a) Display medicines belonging to a particular category.

```
SELECT * FROM medicines
WHERE category = 'Analgesic';
```

(b) Display medicines costing more than ₹50.

```
SELECT * FROM medicines
WHERE price > 50;
```

(c) Display medicines costing less than ₹100.

```
SELECT * FROM medicines
WHERE price < 100;
```

(d) Display medicines having stock less than 30.

```
SELECT * FROM medicines
WHERE stock < 30;
```

(e) Display medicines manufactured by a particular manufacturer.

```
SELECT * FROM medicines
WHERE manufacturer = 'ABC Pharma';
```

(f) Sort medicines according to price.

Ascending Order:

```
SELECT * FROM medicines
ORDER BY price ASC;
```

Descending Order:

```
SELECT * FROM medicines
ORDER BY price DESC;
```

(g) Sort medicines according to stock.

Ascending Order:

```
SELECT * FROM medicines
ORDER BY stock ASC;
```

Descending Order:

```
SELECT * FROM medicines
ORDER BY stock DESC;
```

Q11. Insert Supplier Details

```
CREATE TABLE suppliers (
   supplier_id INT PRIMARY KEY,
   supplier_name VARCHAR(100),
   phone VARCHAR(15),
   city VARCHAR(50)
);

CREATE TABLE medicine_suppliers (
   medicine_id INT,
   supplier_id INT,
   FOREIGN KEY (medicine_id)
   REFERENCES medicines(medicine_id),
   FOREIGN KEY (supplier_id)
   REFERENCES suppliers(supplier_id)
);

INSERT INTO suppliers VALUES
```

```
(1, 'Rahul Traders', '9876543210', 'Roorkee'),
(2, 'Medical Distributors', '9876543211', 'Dehradun'),
(3, 'Health Suppliers', '9876543212', 'Haridwar');

INSERT INTO medicines VALUES
(401, 'Dolo 650', 'Analgesic', 30, 100, 'ABC Pharma'),
(402, 'Azithral', 'Antibiotic', 110, 50, 'HealthCare Ltd'),
(403, 'Vitamin D', 'Supplement', 75, 60, 'Wellness Pharma');

INSERT INTO medicine_suppliers VALUES
(401, 1),
(402, 2),
(403, 3);

SELECT medicines.name, medicines.medicine_id,
      suppliers.supplier_name,
      suppliers.phone, suppliers.city
FROM medicines
INNER JOIN medicine_suppliers
ON medicines.medicine_id = medicine_suppliers.medicine_id
INNER JOIN suppliers
ON medicine_suppliers.supplier_id = suppliers.supplier_id;
```

Q12. Generate Medicine Inventory Report

```
SELECT COUNT(*) AS total_medicines
FROM medicines;

SELECT SUM(stock) AS total_stock
FROM medicines;

SELECT * FROM medicines
WHERE stock < 30;

SELECT * FROM medicines
ORDER BY price DESC
LIMIT 1;

SELECT * FROM medicines
ORDER BY price ASC
LIMIT 1;
```

Q13. Complete Hospital Medicine Management System

```
CREATE DATABASE IF NOT EXISTS HospitalDB;
USE HospitalDB;

CREATE TABLE IF NOT EXISTS medicines (
   medicine_id INT PRIMARY KEY,
   name VARCHAR(100),
   category VARCHAR(50),
   price DECIMAL(10,2),
   stock INT,
   manufacturer VARCHAR(100)
);

INSERT INTO medicines VALUES
(501, 'Paracetamol', 'Analgesic', 25, 100, 'ABC Pharma'),
(502, 'Amoxicillin', 'Antibiotic', 80, 50, 'HealthCare Ltd'),
(503, 'Cetirizine', 'Antihistamine', 35, 20, 'ABC Pharma'),
(504, 'Omeprazole', 'Antacid', 60, 15, 'MediLife');
```

```
SELECT * FROM medicines;

INSERT INTO medicines VALUES
(505, 'Vitamin C', 'Supplement', 45, 80, 'Wellness Pharma');

SELECT * FROM medicines
WHERE medicine_id = 501;

UPDATE medicines
SET stock = 150
WHERE medicine_id = 501;

DELETE FROM medicines
WHERE medicine_id = 504;

SELECT * FROM medicines
WHERE stock < 30;

SELECT * FROM medicines
WHERE category = 'Analgesic';

SELECT * FROM medicines
ORDER BY price DESC;

SELECT COUNT(*) AS total_medicines
FROM medicines;

SELECT SUM(stock) AS total_stock
FROM medicines;

SELECT * FROM medicines
ORDER BY price DESC
LIMIT 1;

SELECT * FROM medicines
ORDER BY price ASC
LIMIT 1;

SELECT * FROM medicines;
```

Result: Hospital Medicine Management System operations such as inserting, viewing, searching, updating, deleting, filtering, sorting and generating inventory reports are performed using MySQL.